

Larkfield Systems

Ninety-one percent of the value was in a number that described the year 2031 and every year after it, and the investment committee had spent forty minutes on the first five years.

Larkfield Systems sold workflow software to hospital pharmacies. Revenue was \$340 million growing at just under 40 percent, gross margin was 78 percent, and the company was roughly at breakeven on an operating basis after five years of heavy spending on its direct sales force. Winterbourne Growth Equity was considering a \$200 million secondary purchase of shares from early investors, and Sofia Amrani had been asked to bring a valuation to the Thursday committee.

Amrani's model was conventional. Five years of explicit forecast: free cash flow to the firm of negative \$20 million in year one, then \$5 million, \$45 million, \$95 million, and \$150 million as the sales investment stopped growing and the installed base compounded. A weighted average cost of capital of 9.5 percent. A terminal growth rate of 4 percent, on the reasoning that healthcare IT would outgrow nominal GDP for a long time.

That produced an enterprise value of \$1,983 million, of which \$1,802 million — 90.8 percent — was the present value of the terminal value.

Amrani had flagged the concentration in the memo. The committee's reaction had been to ask for a sensitivity on terminal growth, which she had provided: at 3 percent the value was \$1,691 million and at 5 percent it was \$2,405 million. A two-point range in a number nobody could observe moved the valuation by 42 percent.

The committee chair, Rufus Oyelaran, had asked a different question.

"Four percent forever. What does Larkfield have to spend, forever, to grow at four percent forever? Because I do not see that number anywhere in your model."

Amrani had not put it in the model. The year-five free cash flow of \$150 million was built off an operating margin of 26 percent and capital spending that had fallen to a maintenance level. Growing at 4 percent in perpetuity would require reinvestment, and the amount required depended on the return Larkfield earned on what it reinvested — a number the model never named because the model never needed it. Except in the terminal value, where it was the only thing that mattered.

YEAR	1	2	3	4	5
Free cash flow to the firm	(20)	5	45	95	150
Weighted average cost of capital					9.5%
Terminal growth rate					4.0%
Present value, years 1–5					\$181.5m
Terminal value at year 5					\$2,836m
Present value of terminal value					\$1,802m
Enterprise value					\$1,983m
Net debt					\$0m
Shares outstanding, fully diluted					41.0m

Year 5 forecast detail: revenue \$1,240m, operating margin 26 percent, tax rate 24 percent, capital spending and capitalized software equal to depreciation.

ASSIGNMENT

- 1 Verify the enterprise value and the share of it that sits in the terminal value. Recompute at terminal growth of 3 and 5 percent.
- 2 Answer Oyelaran's question. If Larkfield grows at 4 percent forever, what fraction of its after-tax operating profit must it reinvest, at a return on invested capital of 12 percent, of 20 percent, and of 30 percent?
- 3 Amrani's year-5 free cash flow assumes capital spending equal to depreciation, which is a reinvestment rate near zero. Reconcile that with your answer to question 2 and rebuild the terminal value consistently.
- 4 What exit multiple of year-5 EBITDA does Amrani's terminal value imply? Sanity-check it against what a mature healthcare software business trades for.
- 5 Winterbourne is buying at \$200 million for a stake priced off a \$1.9 billion enterprise value. State the one assumption in this model that, if wrong, costs Winterbourne the most, and say how you would test it before Thursday.